

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What is drying. with neat & clean diagram, write about spray dryer.
- Q.24 Define HTU and NTU and their significance in packed columns.
- Q.25 Write short note on any two:
- a) dry bulb & wet bulb.
 - b) flooding & channeling
 - c) Humidity chart
 - d) HTU & NTU.

No. of Printed Pages : 4

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Roll No.

4th Sem / Chemical (Pulp & Paper)

Subject : Mass Transfer

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which has the lowest value of diffusivity?

- a) Glucose b) Ethanol
- c) Glycerin d) Water

Q.2 What is the meaning of steady-state diffusion?

- a) No diffusion occurs
- b) Diffusion rate is zero
- c) Concentration remains constant with time
- d) Concentration changes over time

Q.3 What does Fick's first law describe?

- a) Diffusion transfer b) Mass diffusion rate
- c) Heat transfer d) Energy transfer

- Q.4 Humidification is :
- Mass transfer operation
 - Heat transfer operation
 - Simultaneous heat and mass transfer
 - None of these
- Q.5 By difference in temperature, head added or removed is:
- Sensible heat
 - Latent heat
 - Heat of vaporization
 - None of these
- Q.6 The humidity is represented in:
- Humidity chart
 - Psychrometric chart
 - None of these
 - Both humidity and psychrometric chart

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Fick's law of diffusion.
- Q.8 What is the concept of packing height in packed towers?

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- Q.9 What is the unit of mass transfer coefficient?
- Q.10 Define dehumidification.
- Q.11 Explain the concept of weeping of columns.
- Q.12 Expand HETP.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain Henry's law.
- Q.14 Explain the mechanism of relative and percentage humidity.
- Q.15 Write some names of gas-liquid contact operations.
- Q.16 What is the difference between humidifier and dehumidifier?
- Q.17 Explain rate of drying curve with diagram.
- Q.18 Define moisture content on wet and dry basis.
- Q.19 What is the relation between film and overall mass transfer coefficient?
- Q.20 Explain the concept of forced diffusion and eddy diffusion.
- Q.21 What is the effect of pressure and concentration on diffusion with time?
- Q.22 Explain mass transfer coefficient with units.

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